

Task 2.1

We got a very high value of 0.9998, which implies strong association.

2.2

We added 3 dyadic covariate matrices using the attribute dataset, we use the literacy score of the receiver as variable first, we put it in each column of the matrix which allows for the literacy level to affect the probability of a friendship tie.

Second we created a matrix to see if 2 students have the same gender where we have a value 1 if they are and 0 if they aren't.

Third we use the HISEI value of the sender as variable in the rows of the matrix this allows the HISEI value of the sender to influence the chance of forming a tie

2.3

Hypothesis i

The coefficient is negative but very small so it isn't statistically significant and doesn't support the hypothesis significantly.

Hypothesis ii

The coefficient is negative and statistically significant which indicates that the hypothesis is supported.

Hypothesis iii

The coefficient is negative and not statistically significant showing no support for the hypothesis (only a very small one for the opposite).

2.4

see code

2.5

see table for values

Hypothesis i

The coefficients are all close to zero and can be positive or negative which shows no support for the hypothesis that students with high literacy scores receive fewer nominations.

Hypothesis ii

There are mostly small values positive and negative with very few slightly higher values, this indicates that same gender friendships are more likely in some but not all classrooms.

Hypothesis iii

there are again only very small values both positive and negative showing no evidence that students with higher HISEI scores send more friendship nominations.

```

library(readr)
library(sna)
library(tidyverse)

setwd("C:/Users/kevin/Downloads/Assignment1/Assignment1/Lintner")

classrooms <- c("01", "02", "03", "04", "05", "10", "12")

results <- list()

for (class in classrooms) {

  w1 <- as.matrix(read.csv(paste0(class, "_W1.csv"), header = FALSE, sep = ";"))
  w2 <- as.matrix(read.csv(paste0(class, "_W2.csv"), header = FALSE, sep = ";"))
  attr <- read.csv("attr.csv", sep = ";")

  attr_class <- attr %>% filter(classroomID == as.numeric(class))

  complete_idx <- which(complete.cases(attr_class))
  attr_class <- attr_class[complete_idx, ]
  w1 <- w1[complete_idx, complete_idx]
  w2 <- w2[complete_idx, complete_idx]

  literacy_matrix <- outer(rep(1, nrow(attr_class)), attr_class$literacy_end)
  gender_matrix <- outer(attr_class$gender, attr_class$gender, FUN = "==") * 1
  HISEI_matrix <- outer(attr_class$HISEI, rep(1, nrow(attr_class)))

  model <- netlm(
    w2,
    list(w1, literacy_matrix, gender_matrix, HISEI_matrix),
    nullhyp = "qap",
    reps = 5000
  )

  results[[class]] <- summary(model)$coefficients
}

results

library(dplyr)

summary_df <- bind_rows(
  lapply(names(results), function(class) {
    data.frame(
      classroom = class,
      intercept = results[[class]][1],
      w1 = results[[class]][2],
      literacy = results[[class]][3],
      gender = results[[class]][4],
      HISEI = results[[class]][5]
    )
  })
)

```

```
summary_df
```

```
model2 <- netlm(  
  w2,  
  list(w1, literacy_matrix, gender_matrix, HISEI_matrix),  
  nullhyp = "qap",  
  reps = 5000  
)  
summary(model2)
```

```
model1 <- netlm(w2, w1, nullhyp = "qap", reps = 5000)
```

```
summary(model1)
```

(row)	classroom	intercept	w1	literacy	gender	HISEI
1	01	0.0269	0.9999	-0.0013	0.0209	0.0009
2	02	-0.1872	0.9999	-0.0015	0.044	0.0034
3	03	-0.089	1.0001	0.0019	0.0886	-0.001
4	04	-0.1279	0.9999	0.0016	-0.0557	0.0022
5	05	0.242	0.5777	0.0009	-0.0421	-0.0017
6	10	0.5139	0.9997	-0.0012	-0.1339	-0.003
7	12	-0.1734	0.9999	0.0042	0.0462	0.0023